Wallace’s contribution

Many 19th-century naturalists contributed to biogeographical knowledge, but one of the most significant was British naturalist Alfred Russel Wallace. After reading Philip Sclater’s account of the global distribution of bird species, Wallace set out to do the same for other animals. He examined all the factors known at the time to be relevant, including changes in land bridges and the effects of glaciations. He produced maps to demonstrate how vegetation influenced animal ranges, and he summarized the distribution of all known families of vertebrates.

Wallace then proposed six zoogeographic regions, which are still largely in use today: the Nearctic (North America), Neotropics (South America), Palearctic (Europe, north Africa, and most of Asia), Afrotropics (south of the Sahara), Indomalaya (South and Southeast Asia), and Australasia (Australia, New Guinea, and New Zealand). The dividing line between these last two regions, which runs through Indonesia, is still known as “Wallace’s Line.”

“The unity of nature means the interrelationship of all physical sciences.”

Alexander von Humboldt

Plate tectonics

Wallace also made some remarkable discoveries from the fossil record. For example, he worked out that early rodents had evolved in the Northern Hemisphere, moving via Eurasia into South America. Later, in 1915, German geologist Alfred Wegener proposed the radical idea that the continents of South America and Africa were once connected, which allowed the spread of tapirs and other species.

Wegener understood that the distribution of species was in part a record of geological history. Species colonize new areas as conditions change, and